Homework Set 10 - sample_mflix

Purpose: This week we continue our exploration of the document model. We will focus on the concept of implicit schema as described in class. This tool is powerful and enables the creator of a dataset to be more flexible and responsive when modeling data.

Instructions: Put your solutions for all problems in a google colaboratory notebook. Create a level-1 header for each problem and then present the solution in subsequent cells. The header must contain the problem number followed by "confidence x" where x is an integer from 1 to 10 with 10 representing high confidence. For full credit you must show your work. The submission is made via a link on UVACanvas.

Domain Extras: n/a

Data

This assignment uses sample data from Mongo Atlas. Use `sample_mflix` for all pipeline and focus questions. (Reference DS 2022 Lab 9 for assistance). Use `pymongo` for all problems to interact with your data.

Reminder: For all problems, when interacting with the data, use `pymongo` and show your code.

10.1 Pipeline: Analysis (10ψ)

For the `movie` collection, list all top-level fields that appear along with the percentage of documents containing that field.

10.2 Pipeline: Analysis (10ψ)

For this problem we probe deeper into the documents and go beyond the top level.

- What percentage of documents in `movie` lack `tomatoes.critic`?
- How many awards have the movies in the database won?

10.3 Pipeline: Analysis (10ψ)

There are two collections for movies, `movie` and `embedded_movies`. Why?

10.4 Pipeline: Visualization (10ψ)

Create a histogram of movie runtimes.

Create a histogram of movie release years.

10.5 Focus: Implicit Schema (10ψ)

This is a review question in the mid-term style. Answer in one or two sentences.

- Implicit Schema vs. Explicit Schema

10.6 Focus: Implicit Schema (10ψ)

This question is a scenario. Answer with a paragraph.

Scenario: You are in charge of a team making a movie database using the document model. You have several members of the team and after a few weeks you realize they are using different fields and nesting in the database.

Q: Is this a problem? Explain why and explain the instructions you would give to your team going forward.

10.7 Project 2: (20ψ)

For this section all items are based on your specific project.

Data Creation

(2ψ) Item 1. Paragraph or two explaining the raw data acquisition process (provenance).

(3ψ) Item 2. Code Table showing the code used to create the data, one row per file, with a brief description and link to code in repo

(1ψ) Item 3. Bias Identification description of how bias could be/was introduced in the data collection process

(1ψ) Item 4. Bias Mitigation description of how biases can be handled/quantified/accounted for in analysis

(3ψ) Item 5. Rationale for critical decisions, especially judgement calls, and places that can introduce/mitigate uncertainty

Metadata

(3ψ) Item 1. Implicit Schema Guidelines for document structure

(1ψ) Item 2. Data Table listing all of the tables in the dataset, one line per table, brief description and link to csv file

(3ψ) Item 3. Data Dictionary Table with one row per feature in the data set containing: name, data type, description, example

(3ψ) Item 4. Data Dictionary quantification of uncertainty for numerical features

10.8 Final Prep: (10ψ)

Short answer: use one paragraph to answer the following question.

Q: When working in the relational model why is it important to normalize your data?

10.9 Final Prep: (10ψ)

Short answer: use one paragraph to answer the following question.

Q: Why are biases introduced at data collection difficult or impossible to mitigate?